



# Linguistic Masking and Its Release in Simulated Electric-Acoustic Hearing

Yuting Ding, Xuefei Wang, Fei Chen

Department of Electrical and Electronic Engineering, Southern University of Science and Technology, Shenzhen, China

fchen@sustech.edu.cn

## Abstract

Combined electric-acoustic stimulation (EAS) with a cochlear implant (CI) and a hearing aid can significantly improve CI user's speech-in-noise performance, which is noted as combined stimulation advantage. The present study aimed to investigate this advantage in the context of linguistic masking, i.e., when the target speech and masker were from the same or different languages, and how combined stimulation and different linguistic masker affected linguistic release from masking (LRM). Mandarin sentences were mixed with 2-talker babble maskers spoken in Mandarin, Cantonese or English, processed by noise vocoder models simulating CI or EAS hearing, and presented to normal-hearing listeners to recognize. Experimental results showed the combined stimulation advantage under linguistic masking with all three languages. The EAS condition yielded significant difference between the scores of two LRMs (i.e., English vs. Mandarin, and Cantonese vs. Mandarin). Under linguistic masking with Cantonese, there was significant difference between the scores of CI- and EAS-based LRMs. The findings of the present work provided new insights on the benefits for CI users using combined electric-acoustic stimulation.

**Index Terms:** cochlear implants, combined electric-acoustic stimulation, combined stimulation advantage, linguistic release from masking

## 1. Introduction

Cochlear implants (CIs) represent a significant advancement in auditory rehabilitation for individuals with profound-to-severe hearing impairment, effectively restoring their hearing capabilities [1]. The functioning of CIs involves the extraction of critical acoustic information from speech input, predominantly through the use of multi-channel temporal envelope waveforms. This extracted envelope information is then used to electrically stimulate the auditory nerves, which facilitates sound perception in the brain. Although many users of CIs demonstrate commendable speech comprehension in quiet settings, their ability to perceive speech is greatly compromised in noisy environments. This challenge has been well-documented within the field of CI technology and remains a focal point for ongoing research and development [e.g., 2-3].

Numerous studies have advocated for the implementation of combined electric-acoustic stimulation (EAS) for CI users who maintain residual low-frequency hearing in the implanted ear [e.g., 4-8]. The EAS approach involves the surgical insertion of a short electrode array into the base region of the cochlea, designed to preserve residual low-frequency acoustic hearing (typically from 20 to 60 dB up to 750 Hz), while users often experience severe-to-profound hearing loss at frequencies above 1000 Hz [e.g., 4]. In this configuration, low-frequency

speech information is delivered through a hearing aid (HA), whereas high-frequency speech information is transmitted via the CI. Early studies have demonstrated that the incorporation of low-frequency acoustic information significantly enhances speech perception for CI users, particularly in challenging listening scenarios characterized by background noise [e.g., 6] and in the comprehension of tonal languages [e.g., 5, 8].

Following early investigations into the mechanisms underlying the combined stimulation advantage (i.e., in contrast to CI-only hearing) in noisy environments, the current study examines the impact of linguistic masking in EAS hearing. Research has demonstrated that linguistic dissimilarity between target and masker speech can enhance speech comprehension, a phenomenon referred to as linguistic release from masking (LRM), particularly under challenging signal-to-noise ratio (SNR) conditions [e.g., 9]. LRM has been documented in numerous studies focusing on English [e.g., 10-11] and Mandarin speech perception [e.g., 12] involving normal-hearing (NH) listeners. For instance, study in [11] showed that at an SNR of 0 dB or lower, monolingual English speakers exhibited poorer recognition of English sentences when masked by an English 2-talker babble compared to a Mandarin 2-talker babble. While hearing-impaired listeners generally demonstrate reduced speech-in-noise performance relative to NH listeners, there is a paucity of research investigating LRM in this population [e.g., 13-14]. Viswanathan et al. observed LRM benefits in simulated CI hearing, where NH listeners recognized English speech more effectively when presented with Dutch maskers than with English maskers while listening to vocoded (a method to simulate CI speech processing or CI hearing) English sentences [13]. Lung et al. found that the linguistic characteristics of the masker significantly affected the recognition of vocoded Mandarin sentences, with increased interference noted from English, Cantonese, and Mandarin, aligning with earlier findings from NH listeners [e.g., 12].

The aim of this study is twofold: 1) We investigate the impact of linguistic maskers on speech perception in simulated combined electric-acoustic stimulation. The target language in this study is Mandarin, with three distinct masker languages (i.e., Mandarin, Cantonese, and English) selected based on their varying degrees of linguistic similarity to Mandarin. 2) We observe two types of LRM effects, i.e., Cantonese vs. Mandarin, and English vs. Mandarin. Consequently, we examine how any observed combined stimulation advantage interacts with the performance associated with these LRM effects.

## 2. Methods

### 2.1. Subjects and materials

Nine native Mandarin Chinese listeners (5 males, aged 22 to 25 years) participated in this study. They were undergraduate students from the Southern University of Science and

Technology, and received compensation for their involvement. All participants were bilingual in Mandarin and English but were not proficient in Cantonese understanding or speaking, and exhibited normal hearing levels (i.e., pure tone thresholds at 250–8000 Hz lower than 25 dB HL). The experimental procedures involving human subjects were approved by the Ethical Review Board of the Southern University of Science and Technology.

The target speech material utilized in this study consisted of sentences derived from the Mandarin Hearing in Noise Test (MHINT) database [15]. The MHINT corpus includes 24 lists, each comprising 10 sentences and 10 keywords per sentence, all recorded by a male native Mandarin Chinese speaker at a sampling rate of 16 kHz.

Three types of linguistic maskers were used in this experiment, i.e., Mandarin, Cantonese, and English. For the creation of Mandarin masker, two male native Mandarin speakers read in Mandarin eight sentences extracted from a newspaper in a naturalistic style. Their readings were recorded in a soundproof booth, digitized at a rate of 16 kHz, and standardized for root-mean-square (RMS) level. The eight sentences read by each speaker were concatenated to generate the one-talker masker signal. The two male Cantonese masker signals were created by the same way used in creating the two Mandarin masker signals. The English masker speech was obtained from the TIMIT database [16], comprising eight sentences from each of two male speakers, which were similarly combined to form two one-talker English masker signals. Each of the 6 (=3 languages  $\times$  2 speakers) one-talker masker signals had a duration ranging from 15 to 25 seconds. More details on creating the linguistic masker signal can be found in [12]. Note that this study only utilized masker signal by male speakers. This was because the MHINT speech was recorded with a male speaker, and using male voice in both target and masker speech may mitigate any potential effects arising from gender differences in linguistic masking.

## 2.2. Signal processing

To generate the 2-talker babble masker in each language, a random segment of the masker speech from each speaker was selected, ensuring that their duration matched that of the target MHINT sentence, and subsequently equalized for RMS level before being combined to form the 2-talker signal. The resulting 2-talker babble was then mixed with the target MHINT sentence (the onsets of babble and target sentences were not aligned) at signal-to-masker ratio (SMR) levels of 0 dB and –3 dB. Finally, the linguistically masked MHINT sentence was adjusted to match the RMS value of the original sentence.

This study employed two signal processing conditions for simulated CI and EAS hearing. The first condition simulated 8-band electric-only stimulation, representative of CI speech processing, using a noise vocoder [e.g., 17–18]. Note that vocoder simulation has been extensively utilized to mimic CI hearing in NH subjects. Real CI users exhibit considerable variability due to factors such as the etiology of hearing loss and differences in implant devices. The vocoder approach proves advantageous for evaluating the effects of various processing parameters (e.g., the number of stimulation channels) in CI, independently of patient-related variables [18]. In implementing the vocoder process, the speech signals were initially processed through a pre-emphasis filter (with a first-order high-pass filter with a cutoff frequency of 1200 Hz), and subsequently bandpass-filtered into 8 frequency bands ranging from 80 to 6000 Hz using sixth-order Butterworth filters. The

cutoff frequencies for the channel allocation of these bandpass filters were set at 80, 221, 426, 724, 1158, 1790, 2710, 4050, and 6000 Hz, calculated in accordance with the cochlear frequency-position mapping function [19]. From each frequency band, the envelope was extracted through half-wave rectification followed by low-pass filtering with a cutoff frequency of 200 Hz using a fourth-order Butterworth filter. The resulting envelope waveform was then multiplied by a white noise carrier signal. Each band's output was further band-limited using the corresponding bandpass filter, and all amplitude-modulated white noise signals (with band-limiting processing) were summed to create the noise-vocoded stimulus. Finally, the amplitude of this stimulus was adjusted to match the RMS power of the original input speech signal.

The second signal processing condition simulated combined EAS hearing. To replicate the effects of EAS in individuals with residual hearing below 600 Hz, the speech signals were low-pass filtered to 600 Hz via a sixth-order Butterworth filter. This low-pass stimulus was combined with the upper 5 frequency bands (ranging from 724 to 6000 Hz) derived from the 5-band noise vocoder. Note that the amplitude of the low-pass stimulus was equalized to that of the CI simulation. Additionally, RMS power scaling was conducted relative to the input speech signal. Further details regarding the CI- and EAS-processed stimuli can be found in [20].

## 2.3. Procedure

Prior to participating in the speech recognition experiment, all participants completed a training session. Both the training and testing sessions were conducted in a soundproof room. The test materials were presented to participants monaurally (i.e., right ear) using circumaural headphones at a self-determined comfortable listening level. Participants were informed that they would be listening to processed Mandarin sentences, with each sentence available for playback up to three times.

During the training session, participants listened to 20 vocoded MHINT sentences (masked by Mandarin and English maskers at SMR of 0 dB) while reading the corresponding transcripts to familiarize themselves with the processed speech materials. In the testing session, they were instructed to listen to the vocoded MHINT sentences without reading any transcripts, and repeat or guess (if unsure) the words they heard. The experimenter scored the participants' responses based on the number of words correctly identified in each sentence. The intelligibility score for each condition was calculated as the percentage of words correctly identified in the target MHINT sentences for each list.

Each participant completed a total of 12 conditions: 3 masker languages (Mandarin, Cantonese, and English)  $\times$  2 SMR levels (0 dB and –3 dB)  $\times$  2 vocoder conditions (CI vs. EAS). A unique list of MHINT sentences was used for each condition, and no target sentence was repeated across conditions. The order of the test conditions was randomized across participants. The entire experiment took approximately one hour to complete for each participant, and a 5-minute break was given every 30 minutes during the testing session.

# 3. Results

## 3.1. Effect of linguistic masking on combined stimulation advantage

Figure 1 shows the Mandarin sentence recognition scores for all conditions. The scores were first converted to rational arcsine

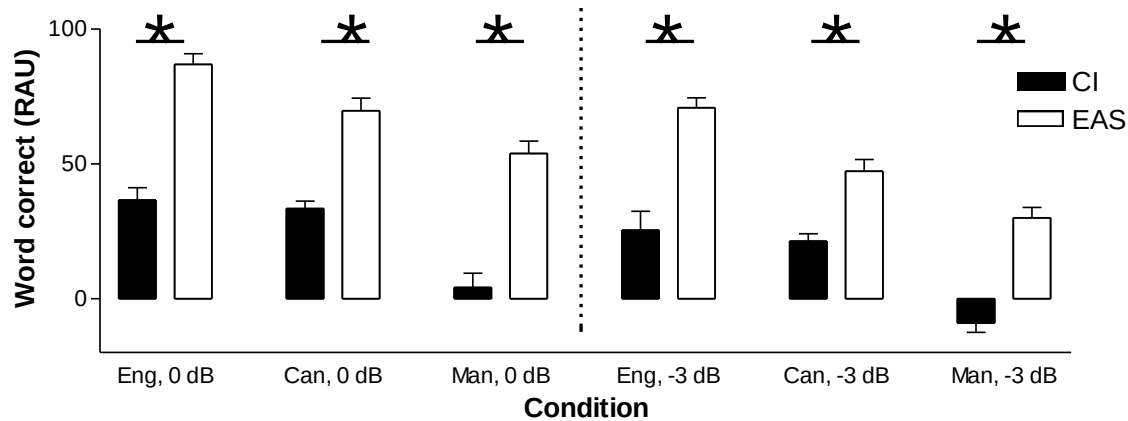


Figure 1: Sentence recognition scores for all conditions. The error bars:  $\pm 1$  standard error of the mean. Asterisk: the difference between the recognition scores of two paired conditions (different vocoder conditions) is statistically significant.

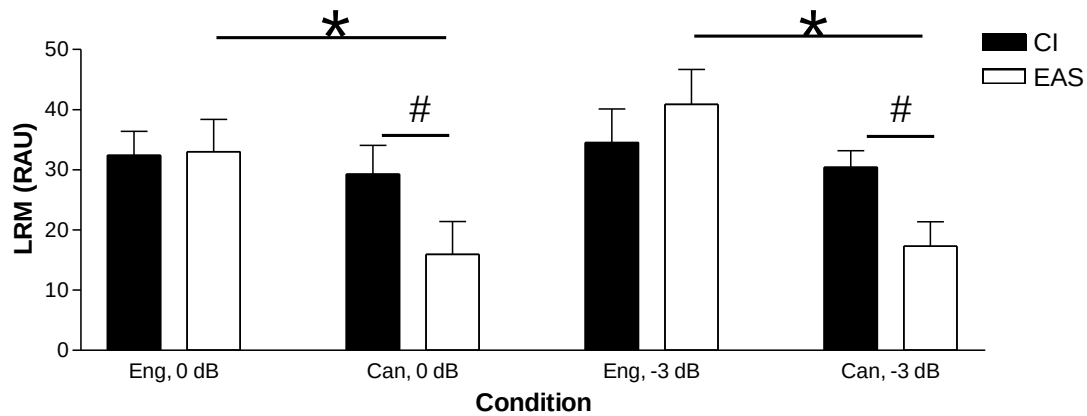


Figure 2: Scores of linguistic release from masking for all conditions. The error bars:  $\pm 1$  standard error of the mean. Asterisk, #: the difference between the LRM amounts the paired conditions is statistically significant.

units (RAU) using the rationalized arcsine transform [22]. The linguistic babble maskers with varying linguistic contents were labeled as “Eng,” “Can,” and “Man,” representing masking by English, Cantonese, and Mandarin speakers, respectively.

Statistical significance was determined by using the recognition score (in RAU) as the dependent variable, SMR level (0 dB vs. -3 dB), masker language (English, Cantonese vs. Mandarin), and vocoder condition (CI vs. EAS) as the three within-subject factors. Three-way analysis of variance (ANOVA) with repeated measures indicated a significant effect ( $F[1, 8]=76.27, p<0.0005$ ) of SMR level, a significant effect ( $F[2, 16]=565.10, p<0.0005$ ) of masker language, and a significant effect ( $F[1, 8]=119.05, p<0.0005$ ) of vocoder condition. Analysis also revealed a non-significant interaction ( $F[2, 16]=0.524, p=0.61$ ) between SMR level and masker language, a non-significant interaction ( $F[1, 8]=3.79, p=0.09$ ) between SMR level and vocoder condition, a significant interaction ( $F[2, 16]=5.47, p<0.05$ ) between masker language and vocoder condition, and a non-significant interaction ( $F[2, 16]=0.13, p=0.88$ ) among SMR level, masker language and vocoder condition.

Post-hoc analysis (with Bonferroni correction if needed) showed that between two vocoder conditions (i.e., CI vs. EAS), all scores at the EAS condition were significantly ( $p<0.05$ ) larger than those at the CI condition, which showed the combined stimulation advantage as expected. With respect to the effect of SMR level, all recognition scores were significantly ( $p<0.05$ ) larger at 0 dB SMR than at -3 dB SMR, except in one pair of scores at the [CI, Eng] condition (i.e., the vocoder condition was CI, and the masker language was English). With respect to the effect of masker language, Mandarin sentence recognition score was significantly ( $p<0.017$ ) larger in English or Cantonese masker than in Mandarin masker under all paired conditions, and significantly ( $p<0.017$ ) larger in English masker than in Cantonese masker, except in the two pairs of scores at the [CI, 0 dB] and [CI, -3 dB] conditions.

### 3.2. Effects of linguistic masking and combined stimulation on linguistic release from masking

With the results of linguistic masking in Fig. 1, two types of linguistic release from masking were determined, i.e., English vs. Mandarin, and Cantonese vs. Mandarin, which was

computed as the difference of recognition scores between two linguistic masking conditions.

Figure 2 shows the LRM scores for all conditions. Statistical significance was determined by using the LRM score as the dependent variable, SMR level (0 dB vs. -3 dB), masker language (English vs. Cantonese), and vocoder condition (CI vs. EAS) as the three within-subject factors. Three-way ANOVA with repeated measures indicated a non-significant effect ( $F[1, 8]=0.93$ ,  $p=0.36$ ) of SMR level, a significant effect ( $F[1, 8]=40.12$ ,  $p<0.0005$ ) of masker language, and a non-significant effect ( $F[1, 8]=1.91$ ,  $p=0.21$ ) of vocoder condition. Analysis also revealed a non-significant interaction ( $F[1, 8]=0.59$ ,  $p=0.46$ ) between SMR level and masker language, a non-significant interaction ( $F[1, 8]=0.07$ ,  $p=0.80$ ) between SMR level and vocoder condition, a significant interaction ( $F[1, 8]=11.17$ ,  $p<0.05$ ) between masker language and vocoder condition, and a non-significant interaction ( $F[1, 8]=0.25$ ,  $p=0.63$ ) among SMR level, masker language, and vocoder condition.

Post-hoc analysis showed that with respect to the effect of SMR level, there was no significant ( $p>0.05$ ) difference between two LRM scores at the conditions with the same vocoder condition and masker language. With respect to the effect of masker language, only the two pairs of LRM scores of EAS hearing at 0 and -3 dB SMR showed significant ( $p<0.05$ ) difference between paired LRM scores, as marked by asterisks in Fig. 2. With respect to the effect of vocoder condition, only the two pairs of scores of Cantonese masking at 0 and -3 dB SMR showed significant ( $p<0.05$ ) difference between paired LRM scores, as marked by '#' in Fig. 2.

#### 4. Discussion and Conclusion

Linguistic masking and its release have been studied for normal hearing and electric hearing (by CI listeners). The present work further investigated how different linguistic maskers affected speech understanding in simulated electric-acoustic hearing (simulating those listeners fitted with CI and hearing aid). First, Figure 1 shows the combined stimulation advantage under conditions with all three linguistic maskers, suggesting that the impact of linguistic masking does not change the established combined stimulation advantage. Hence, the present results further extend the scope of combined stimulation advantage. Second, EAS hearing demonstrates LRM. That is, when there is a large linguistic difference between target and masker speech, listeners could benefit this linguistic release from masking. Figure 1 clearly shows that, in relative to condition of Mandarin target and Mandarin masker (or linguistically matched condition), the linguistically un-matched condition of Mandarin target and English masker or Mandarin target and Cantonese masker yields improved speech understanding performance. In addition, as the linguistic difference between Mandarin and English (e.g., tonal vs. nontonal) is larger than that of between Mandarin and Cantonese (tonal vs. tonal, and sharing common phonemes), the linguistically un-matched condition of English masker yields larger intelligibility than that of Cantonese masker. This result is consistently with early findings from normal and CI listeners [e.g., 12, 14].

The present further studied how combined stimulation and linguistic masker affected the level of linguistic release from masking. Some interesting results were revealed in these analyses (see Fig. 2). First, it is seen that while LRM with English masking is largely larger than that with Cantonese masking, statistical analysis showed that only under the EAS condition, the difference between two LRMs (English masking

vs. Cantonese masking) was significant ( $p<0.05$ ), while under the CI condition, there was no significant difference between two LRMs. This seemingly added new insights on the advantage of combined stimulation. While the positive LRM effect was also observed in CI hearing, we hypothesize that the limited temporal and spectral cues provided by CI speech processor diminished the effect of un-matched linguistic masking. That is, when masker was English or Cantonese, their masking effects did not make a significant difference when they were processed by CI speech processor or characterized by a limited number of temporal features in CI hearing. In contrast, the usage of hearing aid in EAS hearing provided important acoustic cues for speech understanding [e.g., 4-8], particularly the tonal and voicing information in low frequency. The Cantonese masker caused a stronger masking effect than the English masker. Therefore, under the EAS condition, a significant ( $p<0.05$ ) difference was observed between the scores of two LRMs (i.e., English masking vs. Cantonese masking). Further studies are warranted to examine this hypothesis.

The second interesting finding is that masking language may also affect the level of LRM. It is seen in Fig. 2 that under the English masking, there is no significant difference between the paired LRM scores of CI hearing and EAS hearing. However, under the Cantonese masking, the score of LRM under the CI hearing is significantly ( $p<0.05$ ) larger than that under the EAS hearing. There is a large linguistic dissimilarity between English and Mandarin, and this large linguistic dissimilarity determines the level of LRM, which is not affected by type of stimulation (i.e., CI or EAS). On the other hand, although there is linguistic difference between Cantonese and Mandarin (yielding positive LRM effect), Cantonese shares some tonal and phonetic features with Mandarin, and adding low-frequency cues in EAS hearing exaggerates the masking effect of Cantonese masker, potentially counteracting the positive LRM effect. Therefore, the EAS hearing receives a smaller amount of LRM than the CI hearing.

The present work could be extended to address the following limitations. First, the testing conditions only included 2-talker masker with the same gender as that of the target speech. It is unclear how the gender difference (between masker and target speech) interacts with the combined stimulation advantage and linguistic release from masking, which warrants further studies. Second, in addition to the widely-used 2-talker masker, other types of masking conditions have also been tested in early works, e.g., using 6-talker masker and cafeteria noise. It would be interesting to examine how these masking conditions affect the findings in this study.

In conclusion, the present study examined the effect of linguistic release from masking (using three linguistic maskers of English, Cantonese, and Mandarin) in the context of simulated electric-acoustic stimulation. Consistent with early findings from NH listeners and simulated CI hearing, the present work found that the recognition of EAS-processed Mandarin sentences was significant influenced by linguistic masking, increasing influence from English, Cantonese to Mandarin. This work also analysed how the type of stimulation and masker language affected the linguistic release from masking. Interaction between the type of stimulation and masker language was found. The EAS stimulation yielded significant difference between the scores of two LRMs (i.e., English vs. Mandarin, and Cantonese vs. Mandarin). Under linguistic masking with Cantonese, there was significant difference between the scores of CI- and EAS-based LRMs.

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